

BUS 445

Predicting Credit Default Risk: Driving credit revenue through predictive analytics

BUS 445
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Presented by:
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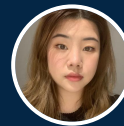
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Executive Summary

Business Problem

High credit default probabilities from risky consumers presents a bank with significant **financial risk**.

Objective

To develop a predictive model that identifies potential high risk customers by assigning them a **calculated odds of default** (risk profile), allowing implementation of **dynamic consumer initiatives** based on risk profiles.

Analytics Problem

Determining the likelihood of credit card default for each bank client based on their profile history.

Which demographic variables are associated with higher default risks?

How does payment and bill status impact default risk?

Are there any observable patterns in defaults by monthly repayment status?

Key Stakeholder

Bank Executives

Credit Card Holders

Credit Requesters





Data and Methodologies

Key Variables and Definitions

Variable	Description	Example	Class	Type
Limit_Bal	Credit limit provided to client	1000	Int	IV
Pay_x Variables	Series of variables representing repayment status	8	Int	IV
Bill_Amtx Variables	Series of variables representing bill statements	9	Int	IV
Sex	Gender	Male (1)	Binary	IV
Education	Encoded representation of education level	2	Int	IV
Marriage	Encoded representation of marriage level	2	Int	IV
Age	Age	49	Int	IV
PAY_AMTx Variables	Series of variables representing amount of payment	1012	Int	IV
Default_next_month	Represents if next month payment defaulted	Yes (1)	Binary	DV



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Dataset consists of 30,000 observations from an Asian retail bank



Methodologies

Objective: To identify the factors contributing to default risk and accurately predict risk

Models:

LOGIT

Purpose:

Find a model that predicts binary outcome of default

Outcome:

Accuracy of +80%

Random Forest

Purpose:

Test collection of models to attempt increase in accuracy

Outcome:

Accuracy of +80%

K Nearest Neighbours

Purpose:

Attempt to classify likelihood of default by analyzing proximity

Outcome:

Accuracy of 61%

Support Vector Machine

Purpose:

Further improve model through alternative classification technique

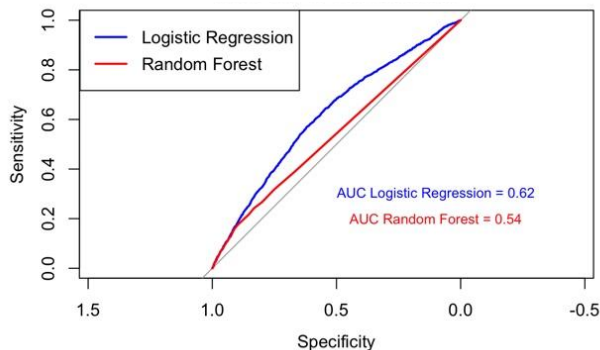
Outcome:

Accuracy: 55%

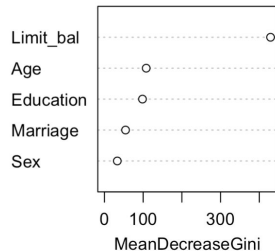
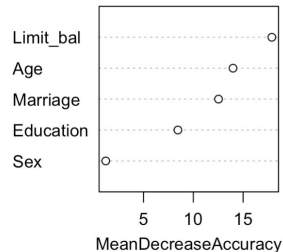
Analysis and Key Findings

Analyzing demographic differences in defaults

AUC Curve Comparison



RF Model

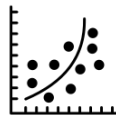


Model Explanation



Limit balance found as the **most significant predictor** for next month's default payment

Accuracy Rates



Logistic regression
77.8%



Random Forest
77.7%



LOGIT: Demographic Analysis

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-5.300e-01	6.212e-02	-8.531	< 2e-16	***
Limit_bal	-3.397e-06	1.633e-07	-20.801	< 2e-16	***
Sex2	-1.763e-01	3.456e-02	-5.102	3.36e-07	***
Education2	3.721e-02	3.992e-02	0.932	0.35129	
Education3	-7.152e-03	5.336e-02	-0.134	0.89338	
Education4	-1.281e+00	4.237e-01	-3.024	0.00249	**
Education5	-1.365e+00	2.797e-01	-4.881	1.05e-06	***
Education6	-4.722e-01	4.539e-01	-1.040	0.29818	
Marriage2	-2.306e-01	3.928e-02	-5.870	4.36e-09	***
Marriage3	-6.606e-02	1.540e-01	-0.429	0.66789	
Marriage4	-1.749e+00	7.315e-01	-2.390	0.01683	*
Age30-39	-3.931e-02	4.437e-02	-0.886	0.37557	
Age40-49	5.278e-02	5.270e-02	1.002	0.31657	
Age50-59	1.182e-01	7.080e-02	1.669	0.09513	.
Age60-69	4.195e-01	1.590e-01	2.638	0.00835	**
Age70-79	-1.669e-01	6.442e-01	-0.259	0.79556	

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Logit model significant at the Limit balance, Sex, Education and Marriage



Key Insights Model 1

Single customers have a lower risk of defaulting

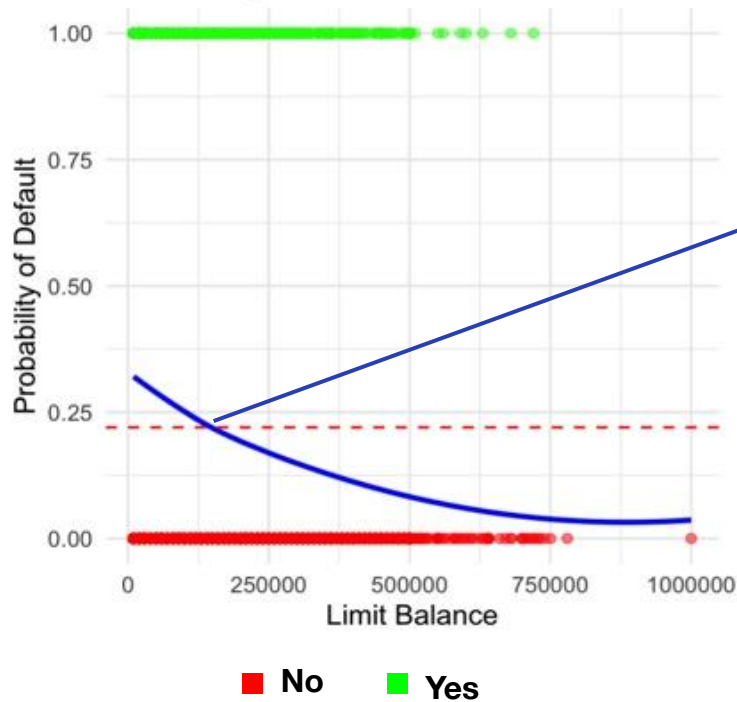
Female has lower default risk than Male

Consumers with a high limit balance are less likely to default



LOGIT: Demographic Analysis

Probability of default based on limit balance



Default risks decreases in a non-linear fashion as limit balance increases

Significant at P-value < 0.001

Bank History - insights

Bank history regression output

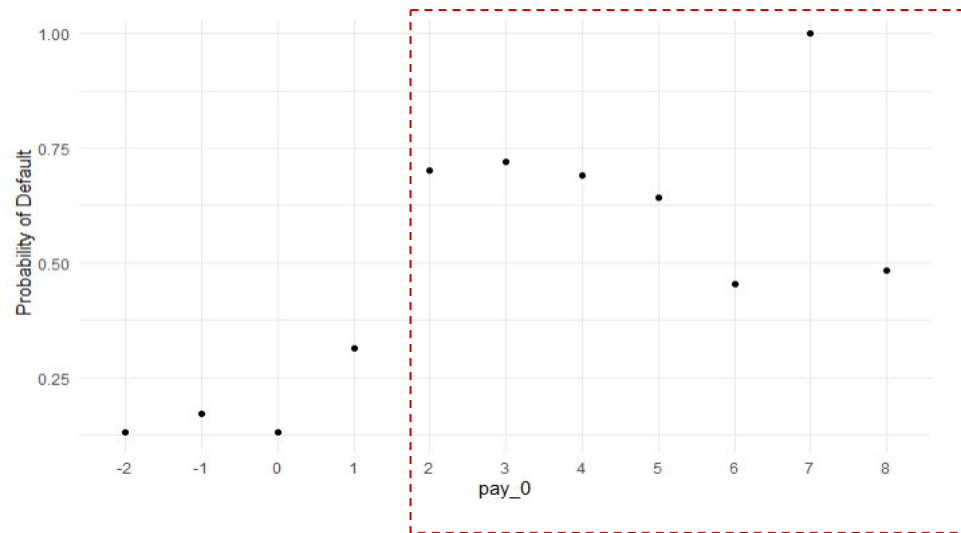
Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.235e+00	9.283e-02	-13.304	< 2e-16	***
PAY_0-1	2.303e-01	9.500e-02	2.425	0.015323	*
PAY_00	-3.493e-01	9.508e-02	-3.674	0.000238	***
PAY_01	7.900e-01	9.781e-02	8.077	6.64e-16	***
PAY_02	2.318e+00	1.074e-01	21.583	< 2e-16	***
PAY_03	2.340e+00	1.951e-01	11.993	< 2e-16	***
PAY_04	2.118e+00	3.596e-01	5.890	3.86e-09	***
PAY_05	1.826e+00	6.356e-01	2.874	0.004056	**
PAY_06	1.123e+00	7.739e-01	1.451	0.146841	
PAY_07	1.383e+01	1.451e+02	0.095	0.924058	
PAY_08	1.236e+00	5.851e-01	2.112	0.034709	*
BILL_AMT1	1.982e-06	4.005e-07	4.948	7.51e-07	***
PAY_AMT1	-1.276e-05	3.033e-06	-4.208	2.58e-05	***
PAY_AMT2	-5.636e-06	2.237e-06	-2.519	0.011770	*
PAY_AMT4	-4.094e-06	2.327e-06	-1.760	0.078452	.
PAY_AMT5	-4.045e-06	2.252e-06	-1.797	0.072409	.
LIMIT_BAL	-2.280e-06	2.232e-07	-10.217	< 2e-16	***

Key Findings

1. Late payments increase default risk
2. Higher pay amounts reduce default risk
3. Higher outstanding bills increase default risk

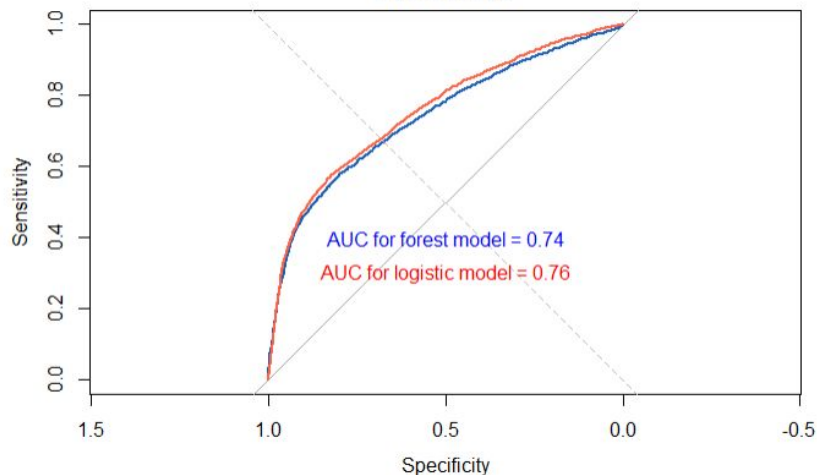
Probability of default based on late payment status



Clients who have previous bills remaining unpaid for greater than 2 months are a greater default risk

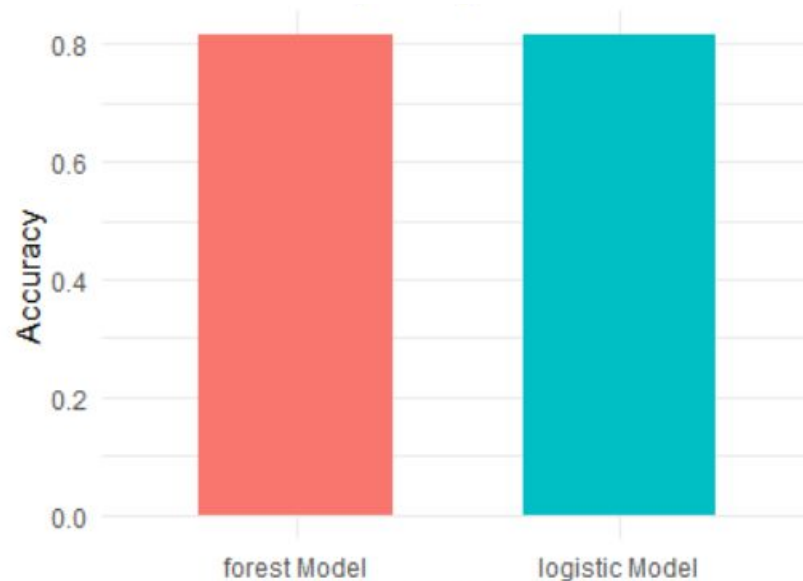
Model performance remains consistent across conditions

ROC Curve



- *Logistic accuracy: 81.8%*
- *Random Forest Accuracy: 81.4%*
- *Bill_Amt1 and Pay_0 found to be the most important predictors*
- **AUC higher than 70% indicating a acceptable fit**

Model Accuracy



~81% accuracy across both applications



Business Value & Managerial Implications

Quantifying Risk Exposure

- Essential for maintaining financial stability
- Develop robust credit risk assessment models
- Offer favourable loan terms to low-risk clients
- Mitigate risk from high-risk clients

Client Segmentation

- Implement proactive risk management strategies:
 - Lower credit limits + monitoring of higher-risk clients
 - Reject high-risk clients from additional credit products
 - Offer favourable loan terms and products to lower-risk clients



Consider two customers, with different default rates...

Attributes	Customer A	Customer B
Pay_0	4	0
Bill Amount	\$500,000 (high)	\$100,000 (low)
Limit Balance	\$300,000 (moderate)	\$800,000 (high)
Marital Status	Single	Married
Gender	Male	Female
Default Probability	70%	10%
Expected Revenue	- \$3,850	\$8,450
Total Expected Revenue	\$4,600	



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***Offering the same interest rate to both customers increases overall risk
→ Adjusting interest rates based on customer risk profiles***





Recommendations

Recommendations

Risk Reduction

Proactive Credit Monitoring

Continuously monitor payment activities of high-risk clients. Implement risk controls based on predictive insights.

Dynamic Interest Rates

Adjust interest rates based on client risk profiles. Offer competitive rates to low-risk clients to increase retention and profitability.

Consumer Retention and Value

Client Segmentation

Offer retention initiatives to low-risk clients. Provide credit education and monitoring for high-risk clients

Upselling Opportunities

Identify and target profitable clients for upselling additional products. Focus on clients with low default probability but high utilization rates





Thank you

Q & A

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Example Calculations

Customer Profiles

Customer 1: PAY_0 category 3, Bill Amount \$300,000, Limit Balance \$500,000, Single Male

Customer 2: PAY_0 category 0, Bill Amount \$100,000, Limit Balance \$800,000, Married Female

Customer A:

Default probability: 70%

Interest revenue if no default: $\$10,000 \times 1.05 = \$10,500$

Cost if default: \$10,000

Expected Revenue for Customer A = $(0.30 \times \$10,500) + (0.70 \times -\$10,000) = (-\$3,850)$

Customer B

Default probability: 10%

Interest revenue if no default: $\$10,000 \times 1.05 = \$10,500$

Cost if default: \$10,000

Expected Revenue for Customer B = $(0.90 \times \$10,500) + (0.10 \times -\$10,000) = \$8,450$

Total Expected Revenue = $(-\$3,850) + \$8,450 = \$4,600$